

Unit 4 : Testing Phase, Product Life Cycle and Value Engineering

Testing is a crucial part of the design thinking process. The testing phase is when the final solution of a product is tested on a full scale. It helps to evaluate whether a solution might or might not work. The prototype that's considered best after considering customer and end-user feedback is implemented. The Testing stage can provide both positive and negative results, depending on the success and failure of the Prototype stage.

Testing might also lead to repeating the Design Thinking process if the end-user faces usability issues with using the solution provided. It provides valuable insights that allow the designer to rework the problem definition at hand and create a new solution that can better empathize with the end-user.

Thus, testing is a repetitive process whereby the prototype must be tested repeatedly to iterate the design.

Why is Testing integral to the Design Thinking Process?

The Testing Phase has an impact on all the previous stages of the Design Thinking Process. Before the Testing Stage is conducted, the envisioned product works on a designer's presumption. While this envisioned product may have sufficient research and empathy behind it, there is a possibility of crucial information being overlooked or of designer bias.

It also provides useful information that aids in human-centered solutions to problems by generating newer product solutions. Conducting repeated testing sessions will give the designer real-time valuable feedback through the observation of real users and not through assumptions.

The Testing stage leads to the iteration of a prototype. It generates user feedback that is particular to a prototype, and this feedback, in turn, deepens the understanding of end-users. Testing helps to bring all the preceding stages together and provide a comprehensive solution that may require adjustments or is ready for large-scale execution.

How is user testing conducted?

The Testing stage, when conducted in an ideal setting (wherein a user is more likely to interact with the product), does give ideal results, but it isn't always possible to do so. Thus, designers can opt to get users to perform a task wherein they're made to interact with the prototype. The idea behind conducting the testing stage on a prototype is to see how end-users might interact with a product in real-time. However, in order to achieve the best possible results of a test, the following steps can be considered -

- **Let end-users analyse multiple prototypes** - By creating multiple prototypes with different variables, end-users are able to compare the prototypes and choose the one that suits their usability needs best. Providing multiple prototypes makes it easier for users to compare and explain why they prefer one prototype over another.
- **Allow end-users to experience the prototypes in real-time** - Observe how users interact with the product. Instead of over-explaining what the user can expect from the product, let the user figure out how to interact with the product. This will show whether the prototype has usability issues or is easy to work with.
- **Encourage users to be vocal through their experience** - As users explore and interact with the prototype, encourage them to talk about what they feel when using the product. This will provide valuable insight into understanding if there are minor inconveniences present in the product that may have gone unnoticed otherwise.
- **Observe while the user interacts with the prototype** - By noticing how the end-user correctly or incorrectly interacts with the prototype, valuable insight can be gained. As this is a process of testing the prototype, observing users in real-time can help to understand how high or low the prototype ranks on usability.
- **Have follow-up queries** - By questioning what the user is trying to say, designers are able to let go of their biases of the product. Even if you believe you've understood what the user is conveying, asking follow-up questions will give better clarity on the issues being faced during interaction with the prototype.

Problem-solving using User Testing:

Testing along with Prototyping is an integral part of the Design Thinking process. User testing enables designers to create user-centric products. By testing the ideas often and from the initial stages, designers are able to identify usability issues that might crop up in case the product is taken to market. Some of the key points in which user testing is a necessity to Design Thinking are -

- **It saves both money and time** - It ensures that the product being released is the most bug-free, user-friendly product possible by discovering faults and usability concerns early on. What happens if the testing step is bypassed in order to develop the product as quickly as possible? After the product is out, designers will have to spend a lot of time and money fixing it. This is not only stressful, but it's also terrible for business.
- **It helps to reveal unexpected insights** - There are always new insights to be discovered, no matter how extensive the initial user research was. Practical usability testing questions will draw attention to issues that would go unnoticed.
- **It helps enhance user satisfaction** - The user always comes first in Design Thinking. By getting first-hand user feedback, designers can make more educated design decisions, thus enhancing user happiness. Being a designer, constant prototyping and testing help keep the focus on the user.

Embracing failure in design thinking:

Embracing failure in design thinking is fundamental to fostering innovation and creative problem-solving. In the iterative process of design thinking, failure is not viewed as a setback but as a valuable source of learning and refinement. Design thinkers understand that experimentation and risk-taking are inherent components of the journey towards finding optimal solutions. By welcoming failure, designers can uncover insights that may not have been apparent initially, leading to more robust and user-centric outcomes.

Failure in design thinking prompts teams to reassess assumptions, iterate on prototypes, and refine their understanding of the problem. It encourages a mindset shift

from fearing failure to celebrating the lessons learned through each iteration. This approach accelerates the design process and cultivates a culture of resilience, adaptability, and continuous improvement within the design team. Embracing failure in design thinking is not an endorsement of mistakes but a recognition that setbacks are integral to the path of innovation and ultimate success.

Testing with End Users:

Including end-user testing is a pivotal stage in the design and development process, holding a key role in ensuring that a product or service effectively resonates with its target audience. Directly engaging end users during testing facilitates the collection of valuable insights into their preferences, pain points, and overall experience. This user-focused approach empowers designers to pinpoint and address potential issues early in the development cycle, mitigating the risk of delivering a product that might fall short of user expectations.

Efficient end-user testing extends beyond assessing mere functionality; it delves into the emotional and experiential dimensions of the user journey. Through observation and feedback collection from real users, designers can refine and optimize the design to better align with user needs and expectations. This iterative process not only elevates the usability and satisfaction of the final product but also instills a user-centered mindset within the design team, underscoring the significance of empathy and continual enhancement in crafting successful and impactful solutions.

Testing with End Users vs Testing without End Users:

The comparison between testing with end users and testing without end users is a pivotal aspect of the design and development process. In testing with end users, direct interaction with the target audience provides insights into preferences, pain points, and the overall user experience, enabling designers to iteratively refine the product to align better with actual user needs. Conversely, testing without end users relies on internal assessments, potentially missing the crucial perspective of the intended users. While internal testing may expedite certain development stages, it risks neglecting user-centric aspects and delivering a product that may not meet user expectations.

The distinction between these approaches underscores the significance of integrating end-user perspectives in both the design and testing phases. Testing with end-users ensures a more accurate representation of real-world usage. It cultivates a user-centric mindset within the design team, ultimately contributing to the development of more successful and impactful solutions.

Iteration:

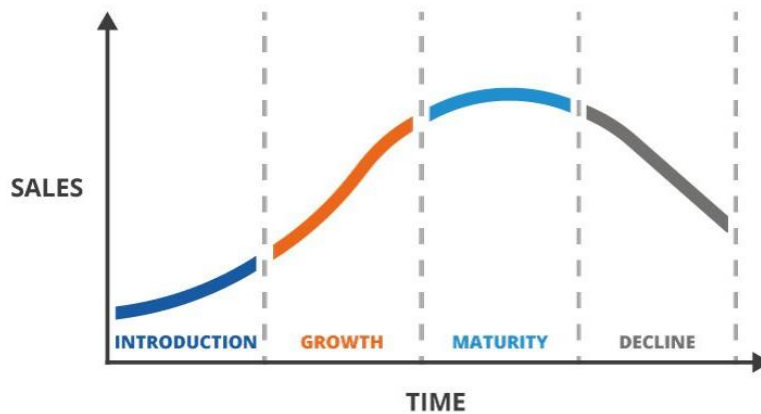
Iteration is a fundamental concept in the design and development process, emphasizing the cyclical and repetitive nature of refining a product or solution. It involves the systematic repetition of stages, allowing for continuous improvement based on feedback and insights gained from each cycle. In design thinking and agile methodologies, iteration is a key strategy to address uncertainties, enhance features, and optimize the user experience. Through multiple iterations, designers and developers can fine-tune elements, make necessary adjustments, and evolve their solutions to better align with user needs and project goals.

The iterative process acknowledges that the initial version of a product is not the final one; rather, it serves as a starting point for ongoing refinement. Iteration promotes adaptability and responsiveness to changing requirements, ensuring that the final output is robust, user-friendly, and aligned with the intended objectives. Embracing iteration in the design and development lifecycle encourages a mindset of continuous learning, flexibility, and improvement, ultimately leading to the creation of more effective and innovative solutions.

Product life cycle: Explain and sketch the different phases in life cycle of the product

A product life cycle is the length of time from a product first being introduced to consumers until it is removed from the market. A product's life cycle is usually broken down into four stages; introduction, growth, maturity, and decline.

PRODUCT LIFE CYCLE



1. **Development:** This initial phase involves the creation and conceptualization of the product. Activities such as research, design, and prototyping are undertaken to prepare the product for market entry.
2. **Introduction:** The product is launched into the market during this stage. Marketing efforts focus on building awareness, and sales typically experience gradual growth.
3. **Growth:** Sales gain momentum during the growth stage, reflecting increased market acceptance. Effective marketing strategies and positive word-of-mouth contribute to the expansion of market share.
4. **Maturity:** The maturity stage signals a stabilization in sales growth. The product has achieved broad acceptance, and companies often concentrate on product differentiation and operational efficiency during this phase.
5. **Decline:** In the decline stage, sales taper off due to market saturation or shifting consumer preferences. Companies may opt to phase out the product or introduce innovations to rekindle interest.

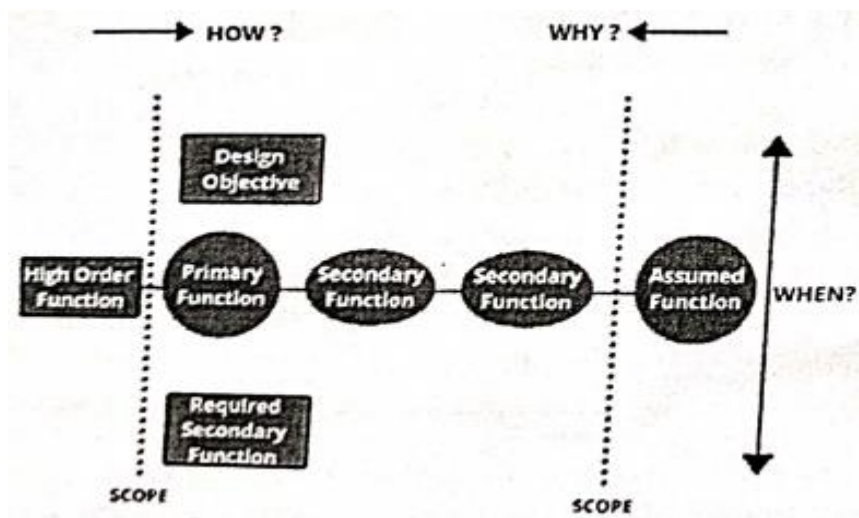
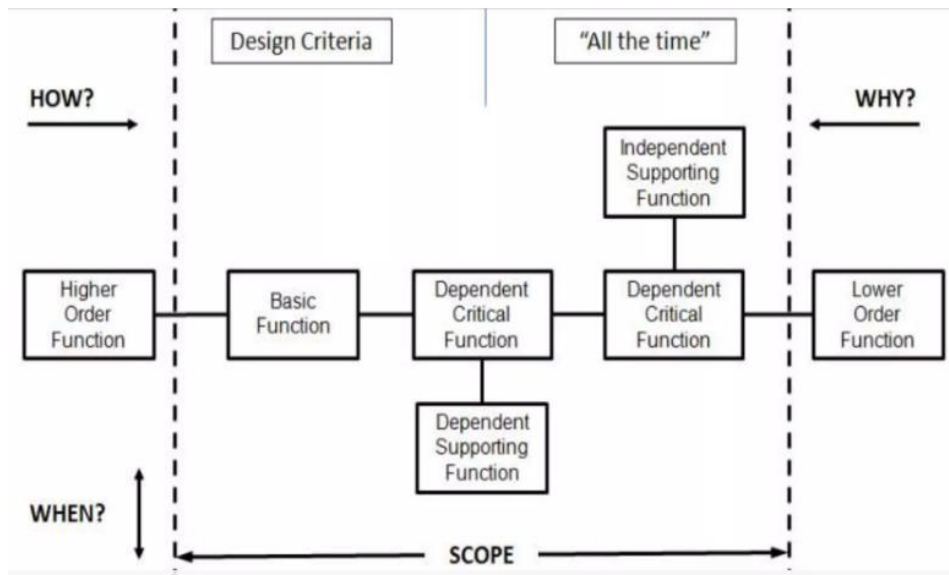
Value Engineering Job Plan

Value engineering, often referred to as value analysis or value management, is a systematic approach that aims to maximize the functionality and performance of a product, system, or process while minimizing costs. The value engineering job plan outlines the step-by-step process to implement value engineering principles effectively.

The value engineering job plan is a strategic and interdisciplinary approach aimed at optimizing project functionality and performance while minimizing costs. It involves defining clear objectives and assembling a diverse team to gather and analyze information. Through functional analysis and creative idea generation, the team identifies core functionalities and explores alternatives. Ideas are evaluated based on their potential to add value, and detailed alternatives are developed with a keen focus on cost analysis and risk assessment. The final stages include developing an implementation plan, gaining stakeholder approval, and executing changes while vigilantly monitoring their impact on project performance and cost. This meticulous process ensures not only efficiency improvements but also significant cost savings without compromising quality, ultimately leading to comprehensive and successful project outcomes.

Question: Explain FAST Diagram: The Function Analysis System Technique and Draw a typical FAST Diagram.

The FAST diagramming tool has been a mainstay with Value Analysis and Value Engineering (VAVE). The FAST diagram visually represents how functions are linked or work together in a system (product or process) to deliver the intended goods or services. By focusing on functions, teams, and individuals can focus on what is truly important and not be constrained by the physical features of products or processes, leading to a better definition of the problem and a clearer path to a solution.



Question : For the given problem, identify the function and general solution. Justify the solution by the concept of value engineering.

Problem 1: Reduce the Guards by Combining Entrance to Strategic Idea

Function: The function in this problem is to ensure security and control access to a strategic idea or area, typically requiring the presence of guards at the entrance.

General Solution: One potential solution is strategically combining entrances, thereby reducing the number of guards needed without compromising security. By

reevaluating the design of access points and consolidating them intelligently, the need for redundant guard positions may be minimized.

Value Engineering Justification: Value engineering supports the idea of optimizing functions while minimizing costs. In this context, the proposed solution aligns with the value engineering principle by maintaining the core function of security while finding a more efficient and cost-effective way to achieve it. Consolidating entrances not only reduces the number of guards required but also streamlines security procedures, potentially improving overall efficiency and resource utilization.

Problem 2: Reduce Travel Time in Your City During Peak Hour

Function: The function here is to facilitate efficient and timely transportation within the city, especially during peak hours, to alleviate traffic congestion and reduce travel time for commuters.

General Solution: A possible solution involves implementing intelligent traffic management systems, introducing public transportation enhancements, and optimizing road infrastructure to ensure smoother traffic flow during peak hours. This may include the use of smart traffic signals, dedicated lanes for public transport, and advanced route planning systems.

Value Engineering Justification: Value engineering emphasizes maximizing performance while minimizing costs. In this scenario, the proposed solution aligns with this principle by addressing the core function of reducing travel time through a comprehensive and efficient approach. By investing in smart infrastructure and transportation systems, the city can achieve a long-term reduction in travel time, optimizing the use of resources and enhancing the overall value for both commuters and the city administration.

Types of testing:

Unit Testing:

Unit testing involves checking small parts of the code, like functions or methods, to ensure they work correctly. For example, a unit test might verify if a specific function accurately calculates the total price of items in a shopping cart.

Integration Testing:

Integration testing verifies that different components or modules of the software work together seamlessly. For instance, it ensures that user inputs from the frontend of a web application are processed correctly by the backend.

System Testing:

System testing evaluates the entire software system to confirm it meets specified requirements and functions as intended. This includes testing overall functionality, features, and performance, ensuring a comprehensive assessment.

Acceptance Testing:

Acceptance testing is the final validation process before deploying the software, involving end-users interacting with the system to confirm it satisfies their requirements and aligns with acceptance criteria.

Regression Testing:

Regression testing ensures that new changes or updates to the software do not negatively impact existing features. It involves re-testing previously examined functionalities to catch unintended side effects.

Performance Testing:

Performance testing assesses how well a software system functions under various conditions, such as high user loads. This helps identify bottlenecks and areas for optimization.

Security Testing:

Security testing aims to identify and rectify vulnerabilities within the software to resist hacking attempts, unauthorized access, and data breaches.

Usability Testing:

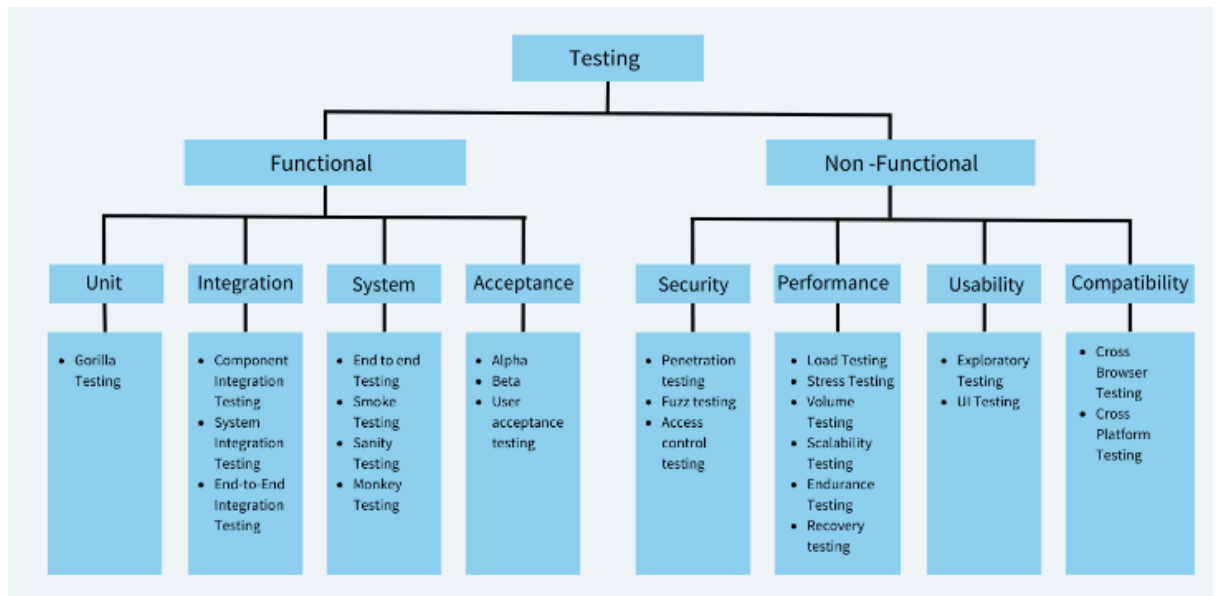
Usability testing evaluates how easy and enjoyable the software is to use from an end-user perspective, providing feedback on the user interface and overall experience.

Compatibility Testing:

Compatibility testing ensures that the software functions across various devices, browsers, and operating systems.

Automated Testing:

Automated testing uses tools and scripts to automatically run tests, making the testing process more efficient and repeatable across various types of testing.



Question scenario: while playing beach tennis on vacation, jems was wondering: " How come these beach ball sets always wear out by the end of vacation? these cheap pallets are good examples of fine value engineering. he gets just enough value for the lowest possible price. He paid Rs 300 for a set that he only intended to use for a few weeks away. Does the above scenario fit in an example of value engineering? consider two takeaways of value analysis and discuss.

Scenario Analysis:

Yes, the scenario described by Jems during his beach tennis game fits as an example of value engineering. Jems is reflecting on the fact that the beach ball sets, particularly the cheap pallets, tend to wear out by the end of the vacation. Despite the limited durability, he acknowledges that these inexpensive sets provide just enough value for the relatively low price he paid – Rs 300. The key element here is the conscious balance between cost and functionality, which aligns with the principles of value engineering.

Takeaway 1: Cost-Effectiveness and Functionality:

One takeaway from this scenario is the emphasis on cost-effectiveness. Jems opts for a beach ball set that serves its purpose adequately for the short duration of his vacation, and the low cost of Rs 300 reflects a deliberate trade-off between affordability and the expected usage period. This exemplifies how value engineering seeks to achieve optimal functionality at the lowest possible cost.

Takeaway 2: Limited Lifespan for Specific Use Case:

Another takeaway is the acknowledgment of a limited lifespan for the product, tailored to a specific use case. Jems is aware that the beach ball set is designed for a short-term, vacation-specific purpose, and he perceives it as a fine example of value engineering. In this context, value engineering considers the intended use and lifespan of the product, aligning features and costs with the specific needs of the user.

In summary, the scenario showcases the application of value engineering principles, where cost-effectiveness and product functionality are carefully balanced to meet the user's specific requirements for a limited duration, reflecting the essence of achieving value for the lowest possible price.